

doors that are unopened are the one you selected and one other door. If you correctly picked the car door to begin with (a 1 in a 1,000 chance), then the host left a random goat door closed. If you picked a goat door (a 999 in a 1,000 chance), the host specifically chose the car door to leave closed. The choice of which doors to open isn't random; the host knows to leave the car door closed. It's almost certain that you didn't pick the car door to begin with, so you should switch to the other door.

Another way to think of it is that you have 1,000 boxes and one box contains a prize. You guess which box the prize is in and the host puts it in your hands. Do you think the prize is in your box or one of the 999 other boxes? You don't need the host to open 998 of the 999 boxes that don't contain a prize; the amount of choice is the same as with the 1,000 doors. The odds that you guessed correctly in the beginning are 1 in 1,000, while the odds that you did not (and that the prize is in one of the other boxes) is a near certain 999 in 1,000.

More information about the Monty Hall Problem can be found at https://en.wikipedia.org/wiki/Monty_Hall_problem.

The Program in Action

When you run *montyhall.py*, the output will look like this:

The Monty Hall Problem, by Al Sweigart al@inventwithpython.com

--snip--

```
+-----+ +-----+ +-----+
|       | |       | |       |
|   1   | |   2   | |   3   |
|       | |       | |       |
+-----+ +-----+ +-----+
```

Pick a door 1, 2, or 3 (or "quit" to stop):

> 1

```
+-----+ +-----+ +-----+
|       | |       | |  ((  | | |
|   1   | |   2   | |   oo  |
|       | |       | |  /_/_|
|       | |       | |  |GOAT||
+-----+ +-----+ +-----+
```

Door 3 contains a goat!

Do you want to swap doors? Y/N

> y

```
+-----+ +-----+ +-----+
|  ((  | |  CAR!  | |  ((  | | | | |
|   oo  | |       | |   oo  |
|  /_/_| |  /_/_| |  /_/_|
|  |GOAT|| |  0  | |  |GOAT||
+-----+ +-----+ +-----+
```